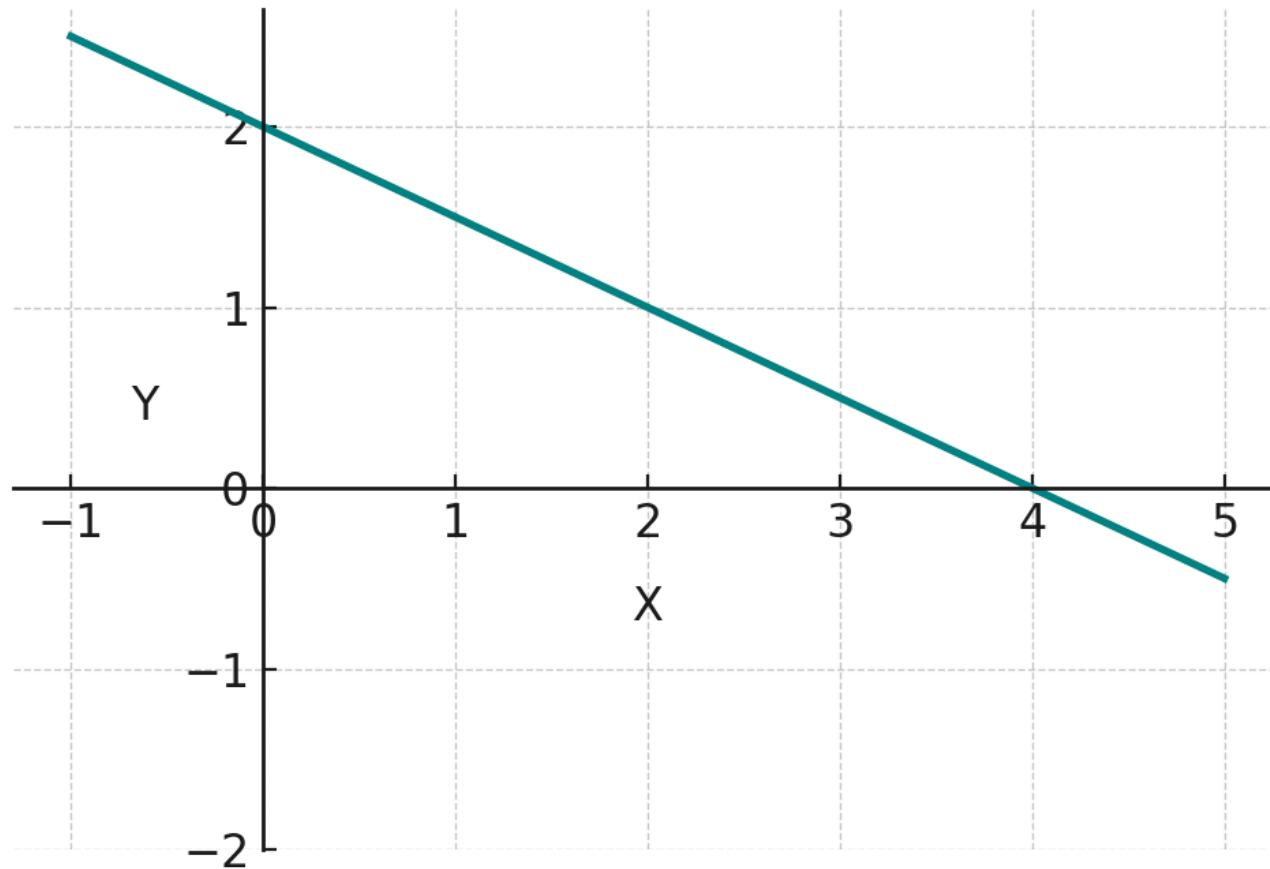


PUBH 526

Topic 1, Part 3:
Regression Review

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Department of Public Health

Equation for a Line

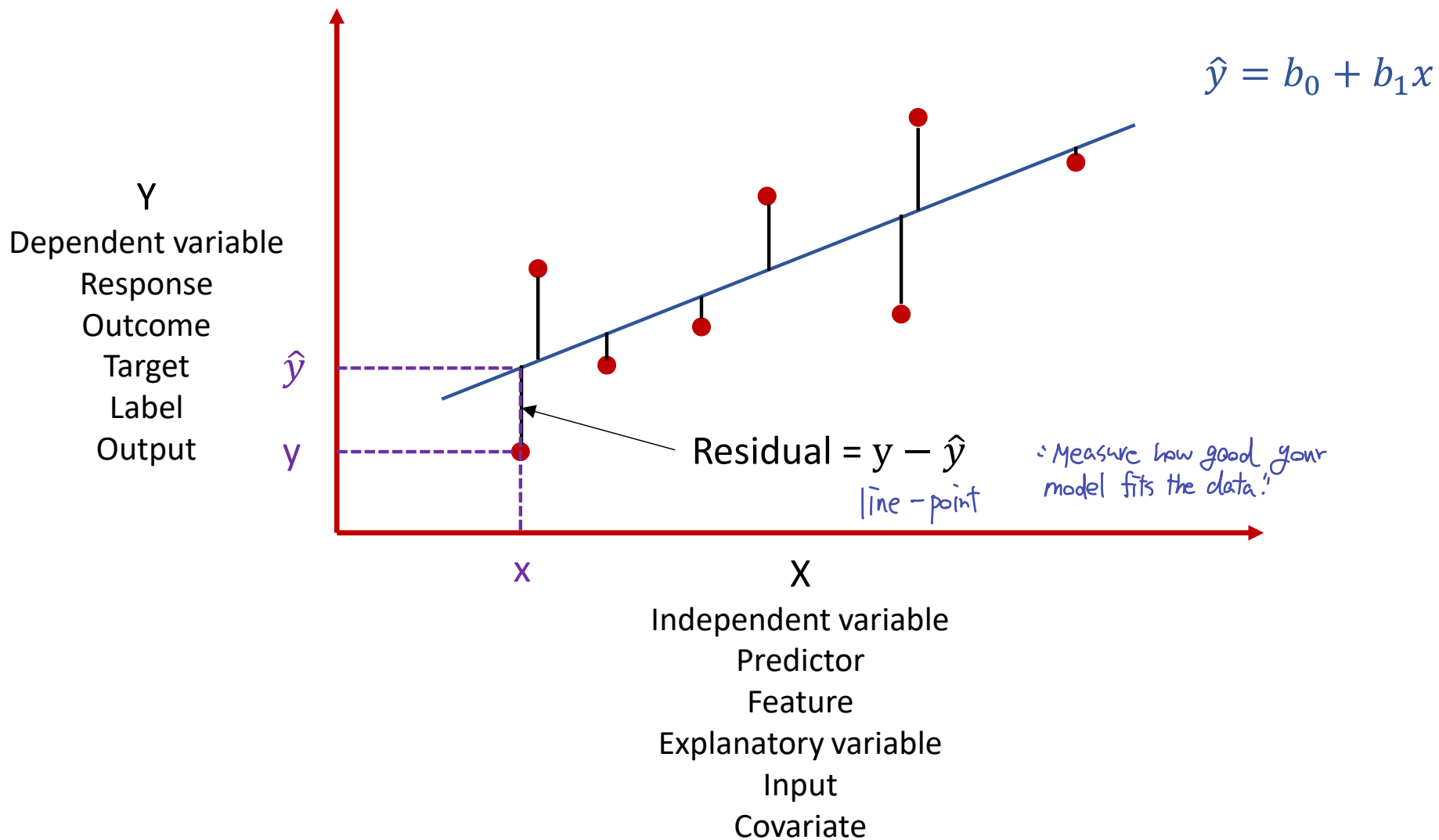


Simple Linear Regression

$\Sigma \text{ residuals} = ?$

$\Sigma (\text{residuals})^2 = ?$
sum of squares

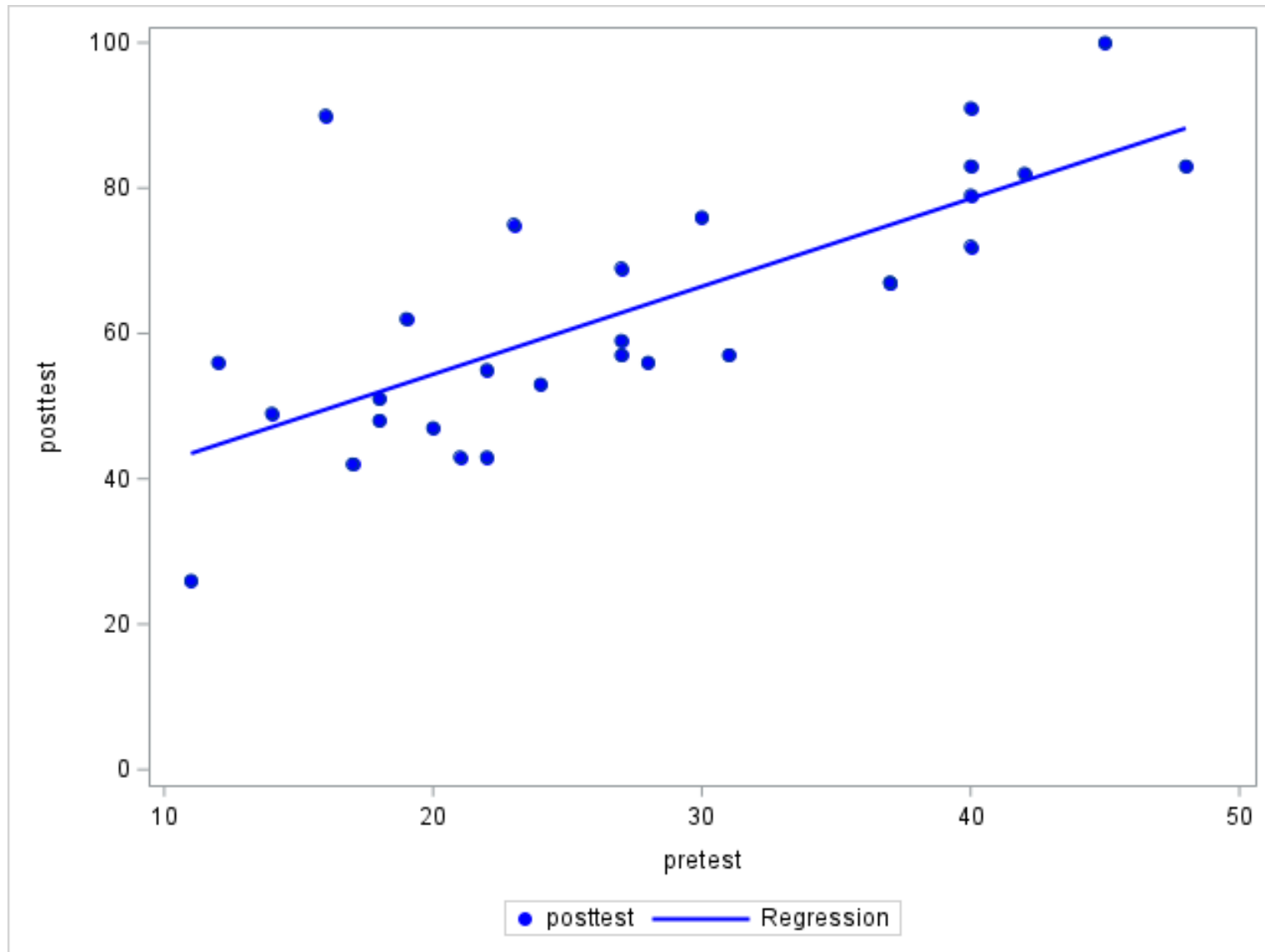
Why do we care about the latter?



Simple Linear Regression

- A local food bank conducted a workshop on healthy meal planning and family nutrition
- All participants took:
 - A pre-test to assess their prior knowledge
 - A post-test to assess their knowledge after attending the workshop
- For each test, the maximum score was 100 points
- The workshop coordinator expected that participants who started out with better scores on the pre-test would do better on the post-test
 - Linear regression: $Y = \text{post-test score}$, $X = \text{pre-test score}$

Always Visualize Your Data



Regression Results

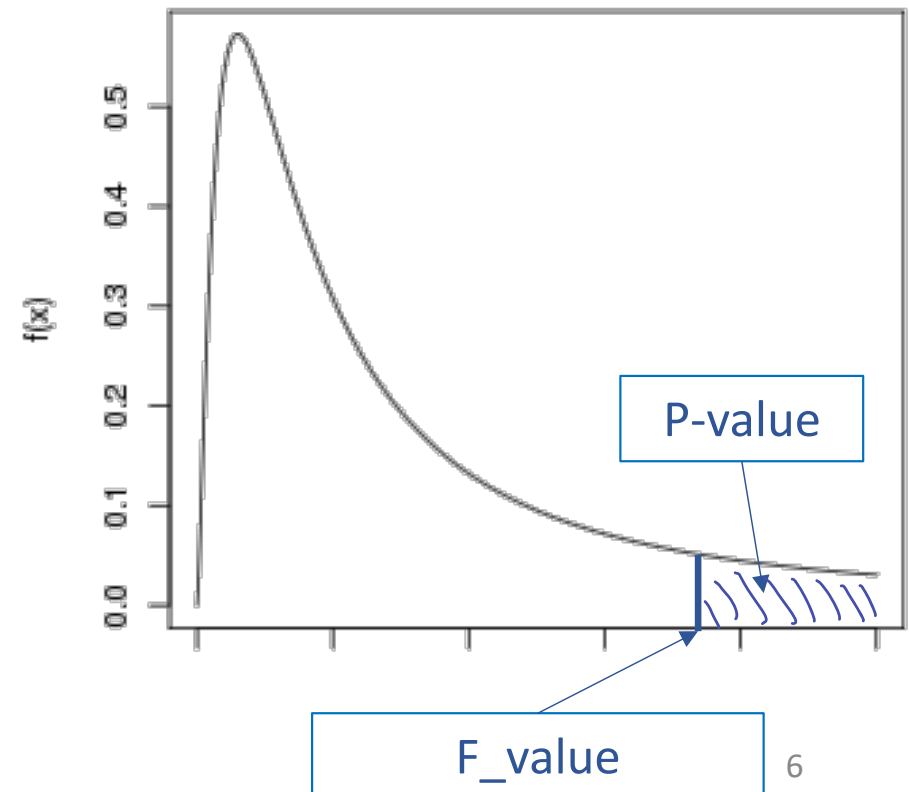
Source	DF	Sum of Squares	Mean Square	F value	Pr > F
Model	1	SSR	MSR = SSR/1	F_value = MSR/MSE	P(F>F_value,1,n-2)
Error	n-2	SSE	MSE = SSE/(n-2)		
Total	n-1	SSTotal			

sample size

$$MS = SS / df$$

$$H_0: \text{slope} = 0$$

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	4594.25136	4594.25136	31.45	<.0001
Error	27	3944.57622	146.09542		
Corrected Total	28	8538.82759			



Regression Results

- How many participants took the workshop? $n=29$
- What is the estimated regression model?

$$\text{Posttest} = 30.2 + 1.2 * \text{pretest}$$

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	4594.25136	4594.25136	31.45	<.0001
Error	27	3944.57622	146.09542		
Corrected Total	28	8538.82759			

Root MSE	12.08699	R-Square	0.5380
Dependent Mean	63.37931	Adj R-Sq	0.5209
Coeff Var	19.07088		

Prop. of variation in y explained by the model

Penalizes the model for adding more predictors: Adjust R^2 based on # of predictors and sample size ($n-p-1$)

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	30.19458	6.32901	4.77	<.0001
pretest	1	1.20899	0.21559	5.61	<.0001

Conclusion?

- Was the coordinator correct that participants who start out with better scores on the pre-test do better on the post-test? How do you know? *Yes, $\beta_1 > 0$
 $\beta_1 = 1.2$, positive $p < 0.05$*
- Did participants actually learn anything? How much did they learn? *Yes. $\beta_0 = 30.2$, even when pretest is still higher than original
 $\beta_1 = 1.2$*
- what does the positive slope actually mean? Do you think the coordinator was happy with this finding? *Yes. For every 1 point $\uparrow$ in pretest, post test $\uparrow$ by 1.2 points.*
- How could the coordinator use this information?
 - Justify effectiveness of workshop*
 - Target future workshops to improve lower-scoring participants.*
 - Show stakeholders that $\uparrow$ engagement, $\uparrow$ baseline knowledge, $\uparrow$ gains.*

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Assumptions

iid: independent and identically distributed

- Observations (participants' scores) are independent
- In fitting the model:

$$\text{posttest}_i = \beta_0 + \beta_1 * \text{pretest}_i + \varepsilon_i, \quad i=1, 2, \dots, 29 \text{ samples}$$

all ε_i follow the same normal distribution with mean 0 and a constant standard deviation, i.e.,

$$\varepsilon_i \sim \text{iid } N(0, \sigma^2) \text{ for all } i$$

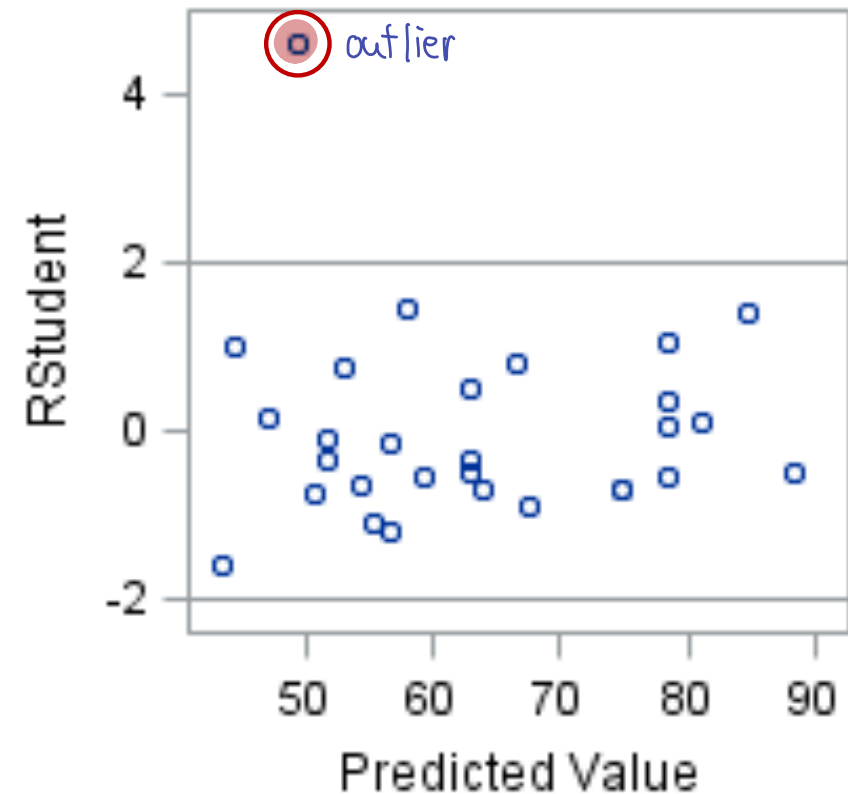
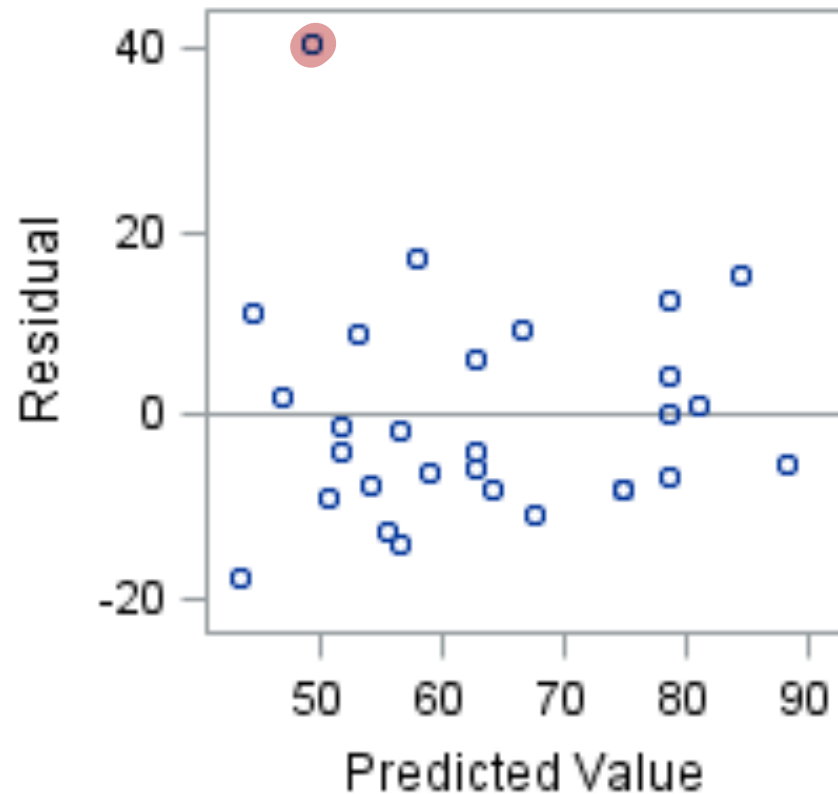
◦ Normality of error terms, not predictors

- Should the independent variable be normally distributed? No
- Should the dependent variable be normally distributed? No

only residuals
need to be normal

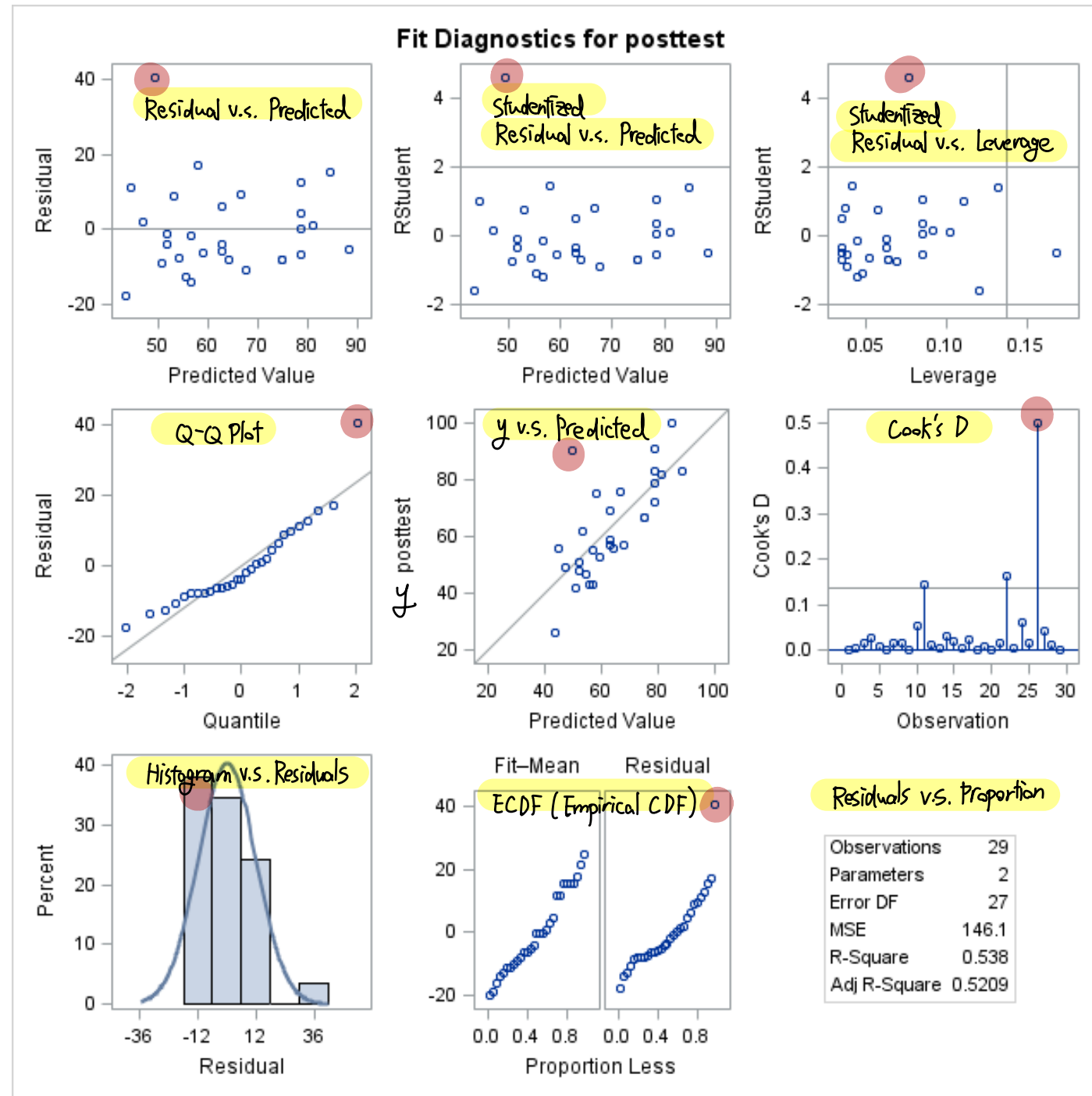
Model Diagnostics

- Residual plots:
 - No particular pattern
 - Linear model probably ok
 - Constant variance



Model Diagnostics

- Potential outlier?
 - Low leverage (not outlying with regard to its X value)
 - High Cook's D (may be influencing our estimated regression line)
 - Return to your plot of the data to explore further
- Residual quantile plot: errors look normal (except perhaps for one outlier)



Hello, SAS...

- “01 regression example.sas”